What's in My Kitchen?

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Task 1: Problem and purpose

College students may have a few ingredients and limited kitchen equipment but still be unsure what they can cook. A recipe can look suitable until they notice a missing ingredient or an oven requirement. This project is a cooking chatbot that helps students understand recipes, check what is missing, and ask simple follow-up questions.

A language model is useful because students can describe their situation in their own words. They can ask which snack uses their ingredients, whether a recipe needs an oven, or how to follow a particular step. A search box can find a recipe, but the chatbot can explain the relevant part and compare it with what the student has. This does not replace a recipe database because retrieval gives the model the recipe information it needs.

The chatbot needs the actual ingredient amounts and instructions in its collection. General cooking knowledge may suggest a reasonable substitution that is not part of the selected recipe. The system therefore uses 1,200 recipe documents and tells the model to avoid inventing ingredients or cooking steps. Success means answering recipe questions correctly, identifying missing ingredients and equipment, remembering earlier messages, and admitting when the collection lacks an answer.

Task 2: How the chatbot works

The application follows my Lab 1 approach. It uses the coursekit chat and embed helpers with the NRP service. Chroma stores recipe embeddings, and a question is embedded to retrieve the closest recipes. The chat model is gpt-oss; the supplied course helper uses qwen3-embedding for embeddings.

Streamlit sends the last six saved messages to the model separately from the recipe context. The current question is added to the saved history after the reply. Memory was checked separately in the interface; the evaluation below uses independent questions with empty histories.

Three prompting techniques: (1) Role prompting assigns the model the role of a cooking helper for college students. (2) Few-shot prompting gives two fictional examples of identifying a missing ingredient and refusing an unsupported question. (3) Structured output asks recipe suggestions to use Recipe, Missing ingredients, Equipment, Steps, and Source recipe sections. Simple questions receive short answers. The examples do not contain answers to the evaluation questions.

Reflection loop: The model writes a draft, then makes one additional call to check it against the same context and return a final answer. The review asks it to remove unsupported claims and correct amounts, steps, and equipment. A separate prompt rule requires a plain refusal when the recipes do not contain the answer.

Task 3: Evaluation set and scoring

The fixed evaluation set contains 20 questions: 17 answerable recipe questions and three questions that should be refused. It covers ingredient matching, amounts, equipment, instructions, and missing information. The full wording and reference answers are in eval.ipynb. The table below summarizes the same questions. I used Microsoft excel for this table.

#	Question summary	Reference answer
1	Fried Onions: oil amount	1 cup vegetable oil.
2	Fried Onions: time and appearance	15-20 minutes; golden brown.
3	Fried Onions: only onions available	Missing vegetable oil.
4	Snack using oats, peanut butter, raisins, maple syrup and cinnamon; no oven	Oatmeal Cookie Energy Bites; mix and roll into balls.
5	Energy Bites: yield	18 bite-size balls.
6	Energy Bites: missing sweetener	Maple syrup.
7	Nutella Grilled Cheese: bread, cheese and butter available	Missing Nutella.
8	Sheet-Pan Grilled Cheese without an oven	Cannot follow as written; needs an oven.
9	Coconut Rice: adding coconut cream	After cooker switches off; pour on top without stirring.
10	Sinangag without eggs	Yes; eggs are optional.
11	Sinangag: oil and garlic for 4 cups rice	1/4 cup garlic oil; 1/2 cup garlic chips.
12	Kateh with American long-grain rice	Do not wash; use 4 1/2 cups water rather than 5 1/2.
13	Big-Batch Rice: reheat without microwave	Saucepan, splash of water, cover, low heat; stir and add water as needed.
14	Almond Butter and Banana Pancakes: missing main ingredient	Almond butter.
15	Brown Butter, Lemon and Parmesan Pasta: sauce too thick	Reserved pasta water, 1-2 tablespoons at a time.
16	Brown Butter Steel-Cut Oatmeal: appliance and cooking time	Instant Pot; 12 minutes at high pressure.
17	Tomato Brown Butter: use raw chunks?	No; puree and strain first.
18	Exact grocery price today	Refuse: current prices are absent.

19	Exact calories per Energy Bite	Refuse: exact calories are absent.
20	Roommate's secret Quantum Moon Noodles	Refuse: private recipe is absent.

Evaluation method

Each of six settings was tested on the same 20 questions three times, producing 360 final answers. The default was temperature 0.2, top-p 0.9, three retrieved recipes, and reflection enabled. Temperature, top-p, and the number of retrieved recipes were changed one at a time. Two additional comparisons disabled retrieval or reflection. All other settings were kept at their default values.

Accuracy is 1 when the essential requested information is correct, with no material invented ingredients or contradictory instructions; otherwise it is 0. Minor extra details affect quality rather than accuracy. For example, a skillet suggestion received a quality penalty, while an invented broth substitution was marked incorrect. These boundary judgments are subjective.

Quality is rated 0 for an unusable answer, 1 for a partly useful answer or unsupported extras, and 2 for a clear, relevant, useful answer. Correct refusals earn accuracy 1 and quality 2. Refusing an answerable question earns accuracy 0. In the no-retrieval condition, a factual answer can still earn accuracy credit against the same reference.

Results

Setting	Mean accuracy (%)	SD (pp)	Quality / 2	Calls / answer
Default	100.00	0.00	1.95	2
Temperature 0.7	100.00	0.00	1.98	2
Top-p 0.5	100.00	0.00	1.97	2
One recipe	98.33	2.89	1.98	2
No retrieval	31.67	2.89	0.53	2
No reflection	100.00	0.00	1.93	1

Every row uses three runs. Mean accuracy is the average of the three run accuracies; SD is the sample standard deviation in percentage points (pp). Quality is averaged over the 60 answers per setting. Calls count chat-generation calls, including reflection, but exclude embedding requests and client retries.

Retrieval mattered most. Accuracy fell from 100% to 31.67% without retrieval, a difference of 68.33 percentage points. The no-retrieval runs scored 30%, 30%, and 35%. The source-only prompt remained in place, so this comparison measures the effect of removing evidence under that policy, not an unrestricted general cooking model.

Temperature and top-p: Neither change affected measured accuracy. One retrieved recipe scored

95%, 100%, and 100%, for a mean of 98.33%. Its 1.67-point difference from default is smaller than its 2.89-point run-to-run SD. Three runs and 20 questions are not enough to conclude that one recipe is reliably worse.

Reflection: Default and no-reflection accuracy were both 100%. Reflection used two calls per answer instead of one. Mean quality was 1.95 versus 1.93, a very small descriptive difference. This test does not show an accuracy benefit from the extra call.

Three diagnosed failures

1. **Unsupported substitution with retrieval.** Setting: One recipe, run 1, question 13. Asked how to reheat frozen Big-Batch Rice without a microwave, the chatbot suggested a splash of water "or a little broth." The retrieved recipe explicitly uses water and does not offer broth as a substitute. The correct source was present, so this is a generation failure rather than a retrieval failure. Reflection did not remove the unsupported suggestion. A better answer would keep the saucepan, water, covered low heat, and occasional stirring exactly as supported by the recipe.
2. **Incorrect coconut cream instruction without retrieval.** Setting: No retrieval, run 1, question
3. The chatbot said to add coconut cream after cooking and stir until evenly distributed. The reference recipe says to pour the cream on top after the cooker switches off and not stir it in. Retrieval was deliberately disabled, leaving the model with no recipe evidence. This is an unsupported generation in the no-retrieval comparison, not a search failure. The model should have refused under its source-only instructions. With the correct recipe supplied, it should follow the no-stir instruction.
4. **Incorrect tomato preparation without retrieval.** Setting: No retrieval, run 2, question 17. The chatbot said to add raw tomato chunks directly to browned butter. The recipe instead requires pureeing and straining the tomato before adding it. Again, retrieval was disabled, so there was no search result to diagnose as wrong. The failure was confidently generating an unsupported instruction and attributing it to a recipe it had not received. Under the existing prompt, it should have refused rather than inventing the method.

What the results do and do not show

The intended source recipe appeared in every retrieved context for the 17 answerable questions when retrieval was enabled. No retrieval-search failure was observed on those questions. The three unsupported-information questions were refused in every setting. This supports the chatbot's performance on this fixed set, but it does not establish reliable refusal on every possible question.

Most questions name the recipe, and several missing ingredients can be guessed from the title. These features make retrieval and answering easier. The 100% scores therefore suggest that this set is too easy to separate temperature and top-p settings. Harder unnamed-recipe requests, ambiguous followups, and conflicting ingredient constraints would be useful future tests, but were not added after the experiments. The evaluation also does not measure memory accuracy or response speed.

Materials used

Course Lab 1 RAG chatbot and the supplied NRP/coursekit helpers; application.py for the chatbot; recipe_data.py for the 1,200-document corpus; eval.ipynb for questions, settings, experiments, and results.